Software Requirements Specification: AI Interview Prep Tool

Oct 13, 2026 · @Someone

1. Introduction

1.1 Purpose

This SRS defines the requirements for an invite-only web app that prepares candidates for interviews for admin-approved job roles. Candidates study flash cards, then complete a voice-enabled AI mock interview that gives feedback after every answer, re-asks the question to check learning, and ends with a preparedness score.

The document is written for the product owner, developers, QA and anyone building or evaluating the first release.

1.2 Scope

The product, referred to here as **PrepPilot** (working name), covers:

- Invite-only access with email-based login
- Admin management of users, job roles, flash cards and question banks
- A flash-card study step per job role
- An AI interviewer that asks questions by voice and text, evaluates answers, gives written and spoken feedback, and re-asks each question once
- Per-question and per-session preparedness scoring, with a results report

1.3 Definitions

Term	Meaning
Candidate	An invited user preparing for an interview
Admin	A user who manages invites, job roles, content and reviews results
Job role	An admin-approved position (e.g. Customer Support Associate) with its own flash cards and questions

Term	Meaning
Flash card	A short study card: a likely question, what a good answer covers, and a tip
Session	One end-to-end mock interview for one job role
Attempt	One candidate answer to one question; each question has a first attempt and a retry
Content feedback	Feedback on what the answer said: correctness, completeness, structure
Delivery feedback	Feedback on how it was said: clarity, pace, filler words, confidence, conciseness
Preparedness score	A 0 to 100 score summarising readiness for the role
LLM	Large language model used as the interviewer and evaluator
STT / TTS	Speech-to-text / text-to-speech

1.4 Document conventions

Requirements use IDs (e.g. FR-AUTH-01). "Shall" marks a mandatory requirement for release 1; "should" marks a desirable one. Priorities are **Must**, **Should** or **Could**.

2. Overall description

2.1 Product perspective

PrepPilot is a new, standalone web app. It relies on three external services: an LLM API for interviewing and evaluation, a speech service for STT and TTS, and an email service for invites and login links. No integration with an ATS or HR system is required in release 1.

2.2 User classes

User class	Description	Key needs
Candidate	Invited job seeker or internal employee; varied tech comfort; mostly on laptop, some on mobile	Simple login, clear steps, honest and actionable feedback, a score they understand

User class	Description	Key needs
Admin	Recruiter, trainer or program manager	Invite users, publish roles and content, see who is prepared and where they struggle
Super admin	Owner of the deployment	Manage admins, AI settings, data retention

2.3 Operating environment

- Latest two versions of Chrome, Edge, Safari and Firefox on desktop; Chrome on Android and Safari on iOS
- Microphone access required for voice answers; a text-only fallback shall exist
- Cloud-hosted; HTTPS only

2.4 Constraints

- Only invited email addresses can log in; there is no public sign-up.
- Candidates see only job roles that an admin has approved and assigned to them.
- AI output must stay within interview coaching; it shall not make hiring decisions.
- Personal data, audio and transcripts must follow applicable privacy law (e.g. GDPR, India's DPDP Act 2023), depending on where users are.

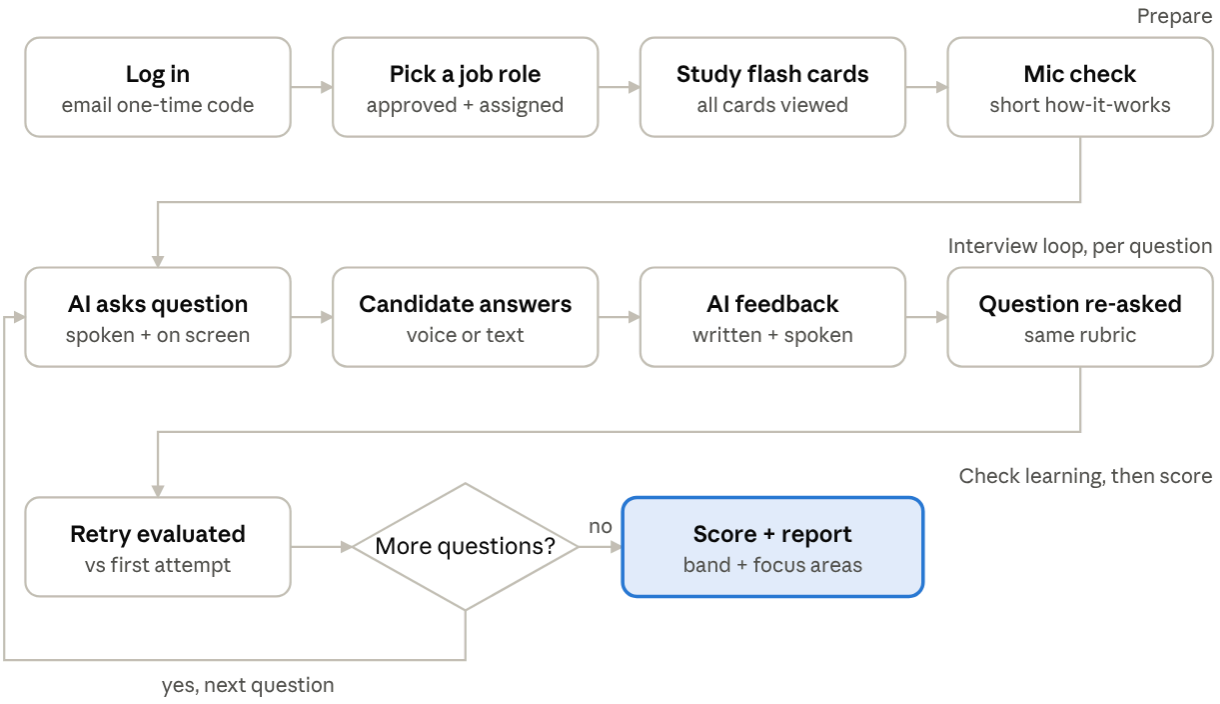
2.5 Assumptions and dependencies

- An LLM can reliably return structured JSON evaluations against a rubric.
- Candidates answer in one language per session (English in release 1).
- Email delivery is reliable enough for magic-link or OTP login.
- Admins supply or approve the flash cards and question bank for each role; AI may draft them for admin review.

3. User journeys

3.1 Candidate journey

Each question is answered, coached, then re-asked before scoring



candidate journey · 11 steps, 1 loop per question

The coaching loop (ask, answer, feedback, re-ask, compare) runs once per question, 5 times by default, before the session is scored.

3.2 Admin journey

1. Log in and create a job role with description, seniority and key skills.
2. Add flash cards and questions with rubrics, or generate drafts with AI and edit them.
3. Preview the role and run a test interview.
4. Approve the role so it becomes visible to assigned candidates.
5. Invite candidates by email or CSV and assign them to the role.
6. Monitor progress and scores; open sessions to review transcripts and feedback.
7. Refine cards, questions and rubrics based on where candidates struggle.

4. Functional requirements

4.1 Authentication and access (AUTH)

ID	Requirement	Priority
FR-AUTH-01	The system shall allow login only for email addresses on the invite list.	Must
FR-AUTH-02	Login shall be passwordless: a one-time code or magic link sent to the email, valid for 10 minutes and single use.	Must
FR-AUTH-03	Uninvited emails shall see a neutral message ("If you're invited, you'll get an email") so the invite list is not exposed.	Must
FR-AUTH-04	Sessions shall expire after 7 days of inactivity; users can log out from any page.	Must
FR-AUTH-05	Login attempts shall be rate-limited (e.g. 5 code requests per email per hour).	Must
FR-AUTH-06	Role-based access: Candidate, Admin, Super admin.	Must

4.2 Admin: users and invites (ADM)

ID	Requirement	Priority
FR-ADM-01	Admins shall invite candidates by single email or CSV upload (email, name, optional roles).	Must
FR-ADM-02	Invites shall send an email with a login link; admins can resend or revoke an invite.	Must
FR-ADM-03	Admins shall assign one or more job roles to each candidate.	Must
FR-ADM-04	Admins shall deactivate a candidate, which ends their sessions immediately.	Must
FR-ADM-05	Admins should set an optional access expiry date and attempt limit per candidate.	Should
FR-ADM-06	All admin actions shall be written to an audit log (who, what, when).	Should

4.3 Admin: job roles and content (ROLE)

ID	Requirement	Priority
FR-ROLE-01	Admins shall create, edit and archive job roles with title, description, seniority and key skills.	Must
FR-ROLE-02	A job role shall have a status: Draft, Approved or Archived. Candidates see only Approved roles assigned to them.	Must
FR-ROLE-03	Admins shall manage flash cards per role: question, what a strong answer covers, sample answer outline, common mistakes, tip.	Must
FR-ROLE-04	Admins shall manage a question bank per role. Each question has text, category (behavioural, situational, role knowledge, motivation), difficulty and an evaluation rubric (expected key points and red flags).	Must
FR-ROLE-05	Admins shall set the number of questions per session (default 5) and whether order is fixed or random.	Must

ID	Requirement	Priority
FR-ROLE-06	Admins should be able to generate draft flash cards and questions with AI from the role description; drafts need admin approval before candidates see them.	Should
FR-ROLE-07	Admins should preview a role exactly as a candidate sees it, including a test interview.	Should

4.4 Candidate dashboard (DASH)

ID	Requirement	Priority
FR-DASH-01	After login, candidates shall see a list of their assigned, approved job roles.	Must
FR-DASH-02	Each role card shall show progress (not started, studying, interview in progress, completed) and the best preparedness score.	Must
FR-DASH-03	Candidates shall be able to resume an unfinished session.	Must
FR-DASH-04	Candidates shall see a history of past sessions with scores and links to reports.	Should

4.5 Flash-card study (CARD)

ID	Requirement	Priority
FR-CARD-01	Selecting a role shall open its flash cards one at a time with next, previous and flip controls.	Must
FR-CARD-02	The front shows the question; the back shows what a strong answer covers, a tip and common mistakes.	Must
FR-CARD-03	The system shall track which cards were viewed.	Must
FR-CARD-04	The "Start interview" button shall appear once all cards are viewed. Admins can make this optional per role.	Should

ID	Requirement	Priority
FR-CARD-05	Candidates should be able to have a card read aloud (TTS).	Could

4.6 AI interview (INT)

ID	Requirement	Priority
FR-INT-01	Before starting, the system shall run a mic check and show a short explainer of how the interview works.	Must
FR-INT-02	The AI interviewer shall introduce itself and the role, then ask questions drawn from the role's question bank.	Must
FR-INT-03	Each question shall be shown as text and spoken aloud via TTS.	Must
FR-INT-04	Candidates shall answer by voice (recorded, then transcribed) or by typing.	Must
FR-INT-05	The candidate shall see a live or post-answer transcript and can re-record before submitting.	Should
FR-INT-06	Each answer shall have a soft time limit (default 2 minutes, admin-configurable) with a visible timer.	Should
FR-INT-07	The interviewer may ask at most one short follow-up when an answer is too vague to evaluate.	Could
FR-INT-08	The AI shall stay in the interviewer persona and decline off-topic requests politely.	Must
FR-INT-09	If the AI or speech service fails, the session state shall be saved and the candidate offered retry or text mode.	Must

4.7 Per-answer feedback (FB)

ID	Requirement	Priority
FR-FB-01	After each answer, the system shall evaluate it against the question's rubric and return feedback within 8 seconds (p90).	Must
FR-FB-02	Written feedback shall include: what was right, what was missing or wrong, and one concrete improvement.	Must
FR-FB-03	Delivery feedback shall cover clarity, structure (e.g. STAR for behavioural questions), conciseness, filler words and confidence of wording.	Must
FR-FB-04	Delivery metrics from voice answers shall include words per minute, filler-word count and answer length.	Should
FR-FB-05	The same feedback shall be spoken aloud in a short summary (about 30 seconds) via TTS.	Must
FR-FB-06	Feedback shall show a sub-score per dimension for the attempt (see section 5).	Must
FR-FB-07	Feedback should include a short example of a stronger answer opening, never a full script to memorise.	Should

4.8 Retry loop (RETRY)

ID	Requirement	Priority
FR-RETRY-01	After feedback, the interviewer shall ask the same question again.	Must
FR-RETRY-02	The retry shall be evaluated with the same rubric, and the evaluator shall also check whether the feedback points were applied.	Must
FR-RETRY-03	After the retry, the system shall show a short comparison: first attempt vs retry, what improved, what still needs work.	Must

ID	Requirement	Priority
FR-RETRY-04	Each question gets exactly one retry in release 1; the number of retries should be admin-configurable later.	Must
FR-RETRY-05	The candidate may skip the retry; a skipped retry counts as no improvement.	Should

4.9 Scoring and results (SCORE)

ID	Requirement	Priority
FR-SCORE-01	At session end, the system shall compute a preparedness score from 0 to 100 and a readiness band (see section 5).	Must
FR-SCORE-02	The results page shall show the overall score, band, per-question scores, improvement between attempts, top 3 strengths and top 3 focus areas.	Must
FR-SCORE-03	Candidates shall be able to replay the transcript and feedback for each question.	Should
FR-SCORE-04	Candidates should be able to download the report as PDF.	Could
FR-SCORE-05	Candidates can retake the interview for a role; each session is scored separately and the best and latest are shown.	Must

4.10 Admin reporting (REP)

ID	Requirement	Priority
FR-REP-01	Admins shall see per-role and per-candidate dashboards: sessions completed, average score, band distribution.	Must
FR-REP-02	Admins shall open any candidate session to see transcripts, AI feedback and scores.	Must
FR-REP-03	Admins should see the questions candidates struggle with most, to improve flash cards.	Should

ID	Requirement	Priority
FR-REP-04	Admins shall export results to CSV.	Should
FR-REP-05	Admins should be able to flag an AI evaluation as wrong, with a note, to improve the rubric.	Should

5. Preparedness scoring model

The preparedness score weights the improved answer most, because the goal is to move each candidate from low to high preparedness, and adds credit for learning between attempts. The first answer still counts a little, as an honest baseline. All weights below are defaults that admins can tune per role.

5.1 Attempt dimensions

The evaluator rates each attempt on four dimensions using a 1 to 5 anchored rubric, mapped to 0, 25, 50, 75 and 100.

Dimension	What it measures	Weight in attempt score
Content	Covers the rubric's key points; accurate; no red flags	45%
Structure	Logical order; STAR or similar for behavioural questions; clear conclusion	20%
Delivery	Clarity, conciseness, filler words, pace (voice), confident wording	20%
Relevance	Answers the question asked; tailored to the role	15%

5.2 Learning score

For each retry, the evaluator lists the improvement points from the first feedback and marks each as applied, partly applied or not applied (1, 0.5, 0). The learning score for that question is the share applied, from 0 to 100. A skipped retry scores 0.

5.3 Session score

```
\text{Preparedness} = 0.20\,\overline{A_1} + 0.55\,\overline{A_2} +
0.25\,\overline{L}
```

Here A1 is the average first-attempt score, A2 the average retry score, and L the average learning score, each from 0 to 100.

5.4 Readiness bands

Score	Band	Message to candidate
85 to 100	Interview ready	Strong and consistent; keep practising delivery
70 to 84	Nearly ready	Solid base; fix the listed focus areas
50 to 69	Developing	Review flash cards and retake
0 to 49	Not ready yet	Study the flash cards again before retaking

5.5 Scoring reliability

- The evaluator shall return structured JSON (scores, evidence quotes from the answer, feedback text) validated against a schema; invalid output is retried once.
- Each score shall cite evidence from the transcript so admins can audit it.
- Evaluation runs at low temperature for consistency. Before launch, scores on a set of at least 30 sample answers per role should be compared with a human reviewer's, aiming for agreement within one rubric level on 80% of answers.

5.6 Progress across sessions

Because the aim is growth, the candidate's dashboard and results page shall show their first-session score as a baseline, their latest score, and the change between them, plus band movement (e.g. Developing to Nearly ready). Admins see the same baseline-to-latest change per candidate and per role.

6. Data model

Nine core entities cover release 1; audio files live in object storage and are referenced by URL.

Entity	Key fields	Relationships
User	id, email, name, role (candidate/admin/super admin), status, access_expires_at, created_at	Has many RoleAssignments, Sessions

Entity	Key fields	Relationships
Invite	id, email, invited_by, token_hash, status (pending/accepted/revoked), sent_at	Belongs to User (admin)
JobRole	id, title, description, seniority, skills, status (draft/approved/archived), questions_per_session, order_mode	Has many FlashCards, Questions
RoleAssignment	user_id, job_role_id, assigned_by, max_attempts	Joins User and JobRole
FlashCard	id, job_role_id, question, strong_answer_points, tip, common_mistakes, order, status	Belongs to JobRole
Question	id, job_role_id, text, category, difficulty, rubric (key_points, red_flags), status	Belongs to JobRole
Session	id, user_id, job_role_id, status, started_at, completed_at, score, band, strengths, focus_areas	Has many Attempts
Attempt	id, session_id, question_id, attempt_no (1 or 2), transcript, audio_url, duration_s, wpm, filler_count, dimension_scores, learning_score, feedback_text, feedback_audio_url, evidence	Belongs to Session and Question
AuditLog	id, actor_id, action, entity, entity_id, timestamp	References User

Card views are stored as a simple CardView record (user_id, flash_card_id, viewed_at). Every LLM call should also be logged with prompt version, model, tokens and latency for cost tracking and debugging.

7. External interfaces

7.1 User interface

The candidate UI has six screens; the admin UI has five.

Area	Screens
Candidate	Login (email + code), Dashboard (assigned roles), Flash cards, Interview setup (mic check), Interview room, Results
Admin	Users and invites, Job roles, Role content editor (cards + questions + rubrics), Reports, Settings

The interview room shows the AI interviewer (avatar or waveform), the current question, a record or type control, a timer, the transcript, and a feedback panel that expands after each answer. Every spoken output also appears as text, so the flow works with sound off.

7.2 AI interface (LLM)

Three prompt roles, each versioned in config rather than code:

1. **Interviewer:** given the role, question and conversation so far, produces the spoken question, transitions and the retry prompt.
2. **Evaluator:** given the question, rubric, transcript and (for retries) the first feedback, returns JSON with dimension scores, evidence, what was right, what was missing, delivery notes, improvement points and a spoken summary.
3. **Session summariser:** given all attempts, returns strengths, focus areas and the band message.

The LLM provider shall sit behind an internal adapter so the model can be changed without app changes.

7.3 Speech interface

- STT: streaming or batch transcription with word timestamps (needed for pace and filler-word metrics).
- TTS: a natural voice for questions and feedback; audio is cached per question to cut cost and latency.

7.4 Email interface

Transactional email (e.g. Amazon SES, SendGrid or Postmark) for invites, login codes and optional session-complete notices. Emails must include SPF/DKIM setup to avoid spam folders.

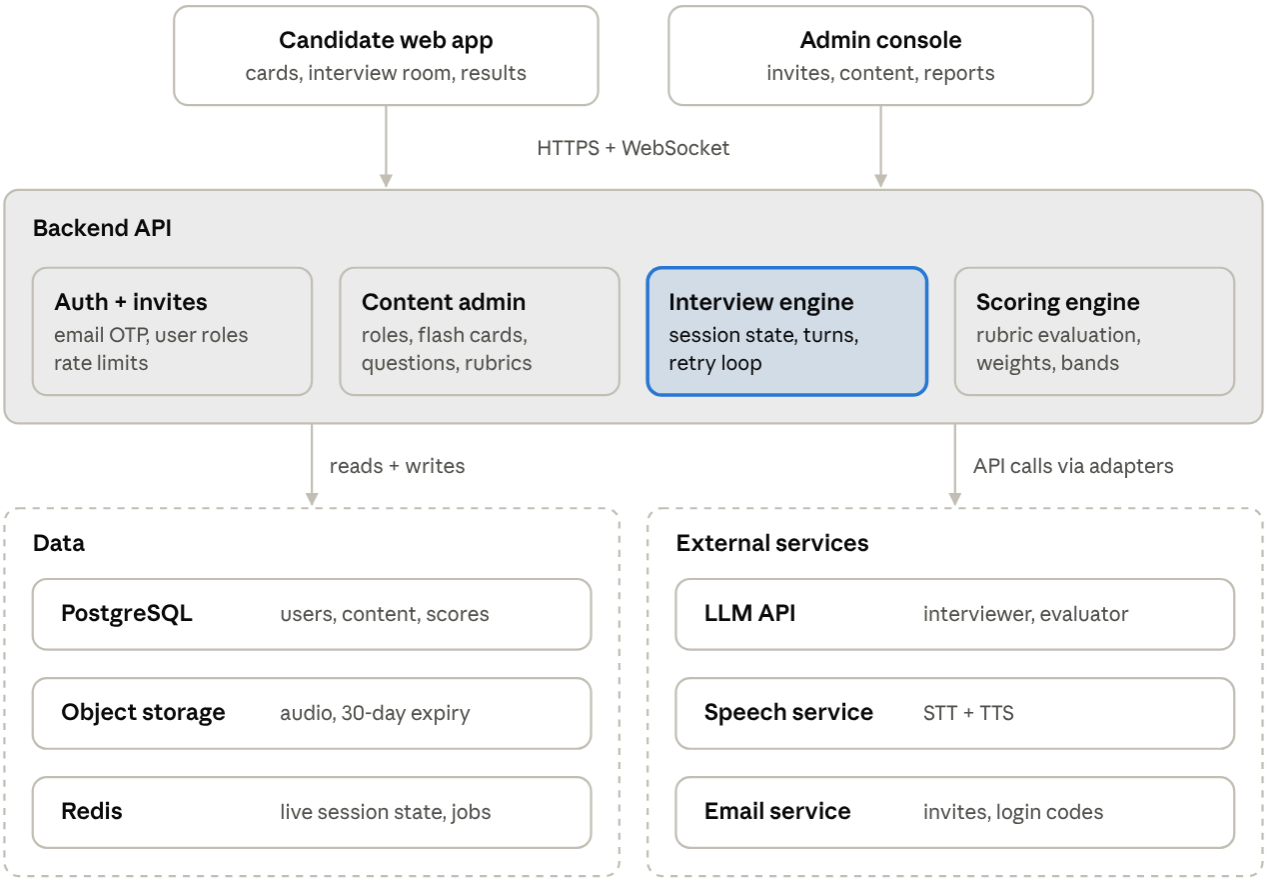
8. Non-functional requirements

ID	Area	Requirement
NFR-01	Performance	Pages load in under 2 s on broadband. Spoken question starts within 2 s. Feedback returns within 8 s at p90.
NFR-02	Scalability	Support 100 concurrent interview sessions in release 1 without degradation.
NFR-03	Availability	99.5% monthly uptime; in-progress sessions survive a page refresh.
NFR-04	Security	HTTPS/TLS 1.2+, encryption at rest, hashed login tokens, OWASP Top 10 controls, admin routes server-side authorised.
NFR-05	AI safety	Prompt-injection guards (candidate text treated as data), output moderation, no hiring decision language, persona limits.
NFR-06	Privacy	Explicit consent before recording audio. Audio deleted after 30 days by default; transcripts kept until the candidate or admin deletes them. Users can request export or deletion.
NFR-07	Accessibility	WCAG 2.1 AA; keyboard navigable; captions for all spoken content; text-only answer mode.
NFR-08	Usability	A first-time candidate can finish login to first question in under 3 minutes without help.
NFR-09	Fairness	Delivery feedback shall not penalise accent or non-native grammar that does not affect clarity; rubrics reviewed for bias.
NFR-10	Cost	Track AI and speech cost per session; target under USD 0.50 per 5-question session (to be validated).
NFR-11	Observability	Structured logs, error tracking, and per-call LLM latency and failure metrics.
NFR-12	Maintainability	Prompts, rubrics and scoring weights are configuration, editable without a code deploy.

9. Suggested architecture and tech stack

A single web backend with four modules is enough for release 1; the interview engine is the core, and every AI, speech and email provider sits behind an adapter.

One backend runs the interview; AI, speech and email are swappable



system architecture · 2 clients, 4 backend modules, 6 services

9.1 One answer, end to end

1. The browser records the answer (MediaRecorder) and uploads it, or sends typed text.
2. The speech service returns a transcript with word timestamps; the engine computes pace and filler-word count.
3. The scoring engine sends question, rubric and transcript to the LLM evaluator and validates the JSON it returns.
4. The attempt is saved; TTS turns the feedback summary into audio.
5. The candidate sees and hears the feedback; the engine then re-asks the question or moves on.

9.2 Suggested stack

Layer	Suggestion	Notes
Frontend	Next.js (React, TypeScript), Tailwind CSS	Responsive; MediaRecorder for audio
Backend	Node.js (NestJS) or Python (FastAPI)	REST plus WebSocket for live interview turns
Database	PostgreSQL	Managed (e.g. AWS RDS or Supabase)
Cache and jobs	Redis	Session state, background jobs
File storage	S3-compatible object storage	Audio with lifecycle expiry
Auth	Passwordless email OTP (e.g. Auth.js, Supabase Auth, or custom)	Invite list checked server-side
LLM	Anthropic Claude, OpenAI or similar	Behind an adapter; JSON-schema output
Speech	STT: Deepgram, AssemblyAI or Whisper. TTS: ElevenLabs, Azure or Google	Word timestamps needed from STT
Email	Amazon SES, Postmark or SendGrid	SPF/DKIM configured
Hosting	Vercel for frontend; AWS or GCP for backend	Region chosen to meet data-residency needs

10. Scope boundaries, open questions and acceptance

10.1 Out of scope for release 1

- Video recording or body-language analysis
- Languages other than English
- Public sign-up, payments or subscriptions
- ATS or HRIS integration, and any use of scores in real hiring decisions
- Native mobile apps (responsive web only)

10.2 Open questions

- [ ] Should candidates see their numeric score, or only the readiness band?

- [ ] Can admins see candidate transcripts by default, or only with candidate consent?
- [ ] Is flash-card completion mandatory before the interview?
- [ ] One admin team for all roles, or separate admins per department or client?
- [ ] Where must data be hosted (e.g. India-only)?
- [ ] Is a talking avatar wanted, or is a voice with a waveform enough?

10.3 Acceptance criteria (release 1)

1. An invited user logs in with an email code; an uninvited email cannot.
2. A candidate sees only approved, assigned roles.
3. A candidate completes all flash cards and starts the interview.
4. For each question, the candidate answers by voice or text and receives written and spoken feedback covering content and delivery.
5. The same question is re-asked, re-evaluated, and a first vs retry comparison is shown.
6. The session ends with a 0 to 100 score, a band, strengths and focus areas that match section 5.
7. An admin can invite users, publish a role with cards and questions, and view every candidate's results.
8. Feedback is returned within 8 s at p90 across 100 test sessions.